Information Systems Security

Learning Objectives

By the end of this lecture, students will be able to:

- Define information security and its importance
- Identify major security threats to information systems
- Explain security goals (CIA Triad)
- Describe common security controls and defense mechanisms
- Understand risk management, disaster recovery, and security policies

Information Systems Security

- Information Systems Security refers to **protecting information and information systems** from:
  - Unauthorized access
  - Misuse
  - Disclosure
  - Modification
  - Destruction
  - Disruption
- Ensure the confidentiality, integrity, and availability of information.

Importance of IS Security

- Protects organizational assets
- Safeguards customer privacy
- Prevents financial loss
- Maintains business continuity
- Complies with legal and regulatory requirements

The CIA Triad

- **1. Confidentiality**
  Prevent unauthorized access to information
  (Encryption, access controls)
- **2. Integrity**
  Ensure data is accurate and not altered
  (Checksums, hashing)
- **3. Availability**
  Ensure information is accessible when needed
  (Backups, redundant systems)
- CIA Triad = foundation of cybersecurity.

Major Threats to Information Systems

- **Threats can be:**
  - **Internal** (employees)
  - **External** (hackers)
- **Common threats:**
- Malware
- Phishing
- Insider misuse
- Denial of Service (DoS)
- Data theft
- Ransomware
- Natural disasters

Malware Types

- **Virus**: attaches to files and spreads
- **Worm**: spreads across networks automatically
- **Trojan horse:** appears legitimate but malicious
- **Spyware**: monitors user activities
- **Ransomware**: locks data until payment

Network Security Attacks

- Common attacks:
  - DoS
  - Man-in-the-middle attack
  - Packet sniffing
  - IP spoofing

Password Attacks

- **Types:**
  - Brute force
  - Dictionary attack
- **Best practices:**
- Strong passwords
- Multi-factor authentication (MFA)
- Regular password changes

Technical Security Controls

- **Firewalls**
  - Control incoming/outgoing network traffic
- **Intrusion Detection Systems (IDS)**
- **Antivirus / Anti-malware software**
- **Encryption** (data at rest & in transit)
- **Access control**
- **VPNs** (secure remote access)
- **Secure configurations**

Encryption

- Encryption protects confidentiality.
- **Types:**
  - **Symmetric encryption** (same key for encryption/decryption)
  - **Asymmetric encryption** (public + private key)
- Examples: AES, RSA, SSL/TLS

Authentication & Access Controls

- Authentication methods:
  - Passwords
  - Smart cards
  - Biometrics
  - Multi-factor authentication

Physical Security

- Protects hardware and physical facilities.
- Controls:
  - Locks and guards
  - Biometric access
  - CCTV
  - Fire suppression systems
  - Environmental controls (humidity, temperature)
  - Backup power (UPS, generators)

Organizational Security Measures

- Security policies
- Security training and awareness
- Incident response plan
- Backup and recovery procedures
- Regular audits and compliance checks

Security Policies

- **A security policy defines:**
  - Acceptable use
  - Password policies
  - Email and internet usage rules
  - Remote access requirements
  - Data classification
  - Backup rules
  - Incident reporting procedures

- Security begins with policy.

Risk Management

- Risk management involves:
  - **Risk Identification**
  - **Risk Analysis**
  - **Risk Evaluation**
  - **Risk Mitigation**
  - **Risk Monitoring**

- Risk = Threat × Vulnerability × Impact

Artificial Intelligence (AI)

Learning Objectives

- By the end of this lecture, students will be able to:
- Define Artificial Intelligence and its goals
- Explain the different types and branches of AI
- Describe major AI applications in organizations
- Understand the difference between AI, Machine Learning, and Deep Learning
- Identify challenges, ethical issues, and future trends

Artificial Intelligence

- Artificial Intelligence (AI) refers to **computer systems designed to perform tasks that normally require human intelligence**, such as:
    - Understanding language
    - Learning from experience
    - Recognizing patterns
    - Making decisions
    - Solving problems

- AI simulates **human thinking and behavior**.

Key Goals of AI

- Automate human tasks
- Improve decision making
- Increase efficiency and accuracy
- Enable machines to learn from data
- Create intelligent solutions for complex problems

Branches of Artificial Intelligence

- Machine Learning
- Deep Learning
- Natural Language Processing (NLP)
- Computer Vision
- Robotics
- Expert Systems
- Speech Recognition

Benefits of AI

- Higher accuracy
- Faster decision-making
- Automation of repetitive tasks
- Reduced human error
- Lower operational costs
- Enhanced customer service

Challenges of AI

- High development cost
- Need for large amounts of data
- Complex implementation
- Skill shortages (AI engineers/data scientists)

Internet of Things (IoT)

Learning Objectives

By the end of this lecture, students will be able to:

- Define the Internet of Things and explain how it works
- Identify the components and architecture of IoT
- Understand IoT benefits

Internet of Things (IoT)

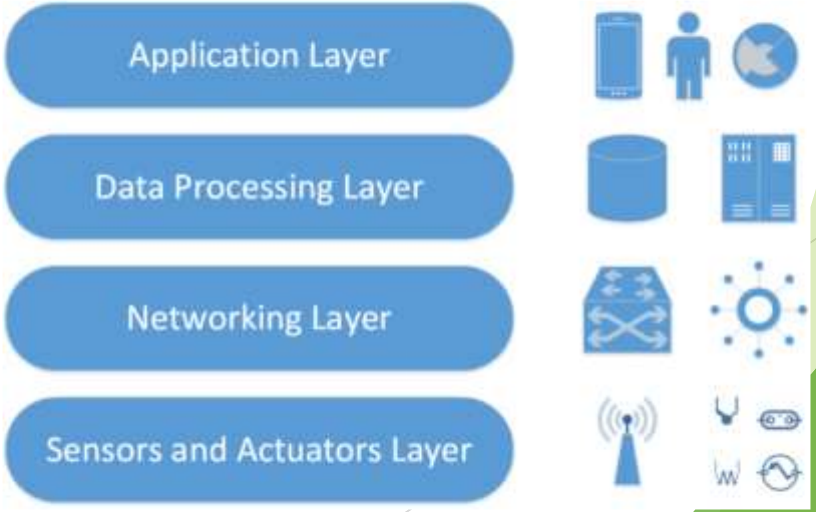
- The **Internet of Things (IoT)** refers to a network of **physical devices** embedded with:
  - Sensors
  - Software
  - Connectivity (Wi-Fi, Bluetooth, cellular, etc.)
- These devices collect and exchange data over the internet with **minimal human intervention.**
- Examples:
  - Smartwatches
  - Smart thermostats
  - Connected cars
  - Industrial sensors

Key Characteristics of IoT

- Connectivity
- Real-time data collection
- Automation
- Remote monitoring and control
- Integration with cloud computing
- Intelligence via analytics and AI

IoT Architecture (4-Layer Model)

- **1 Sensors/Devices Layer**
  - Collects data from the environment
    (e.g., temperature sensors, motion sensors)
- **2 Connectivity/Network Layer**
  - Transfers data to cloud or servers (e.g., Wi-Fi, 5G, RFID, Bluetooth)
- **3 Processing Layer**
  - Data storage and analysis using:
  - Cloud computing
  - Edge computing
  - Big data analytics
- **4 Application Layer**
  - User-facing services such as:
  - Smart homes
  - Smart healthcare
  - Industrial IoT



Components of IoT

- Sensors & Actuators
- Microcontrollers & Edge Devices
- Communication technologies (5G, Wi-Fi, BLE)
- Cloud platforms (AWS IoT, Google IoT Core)
- User interfaces (apps, dashboards)

Example — How IoT Works (Smart Home)

- Sensor detects temperature
- Sends data to the cloud
- Cloud analyzes the data
- Decision-making: room too hot
- Command sent to AC
- AC turns on automatically

Applications of IoT — Smart Homes

- Smart lighting
- Security cameras
- Smart thermostats
- Voice assistants (Alexa, Google Home)

- Benefits: convenience, energy efficiency, automation

IoT and Artificial Intelligence (AIoT)

- AI + IoT = **Smart IoT**
- Examples:
  - Smart cameras that detect faces
  - Self-driving cars
  - Intelligent manufacturing robots
- AI gives IoT:
  - Predictions
  - Pattern recognition
  - Automation

Benefits of IoT

- Automation of daily tasks
- Improved efficiency
- Cost savings
- Real-time monitoring
- Better decision-making
- Reduced human errors